

CS & IT ENGINEERING



Computer Network - 2

Network Layer

Lecture No. - 05

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Recap of Previous Lecture



Topic

Supernetting

Topic

NAT



Topics to be Covered



Topic

NAT

Topic

Traffic Shaping

Topic

DHCP



Topic : NAT



→ NAT : Network Address Translation

→ Internet : Public Network

→ Every network is considered as a private network



Topic : NAT



→ Every **connected network** is identified by unique public IPv4 Address
[Assigned by **ISP**]

→ Total number of network can be exist (world wide) is 2^{32}

→ All **hosts inside a network** is identified by private IPv4 Address



Topic : NAT Device



→ For every outgoing datagram, →
it modify Source IP address from private IP address to public IP address

→ For every incoming datagram, →
it modify Destination IP address from public IP address to private IP address

[MSQ] [GATE-2024, Set-1, 1 Mark]

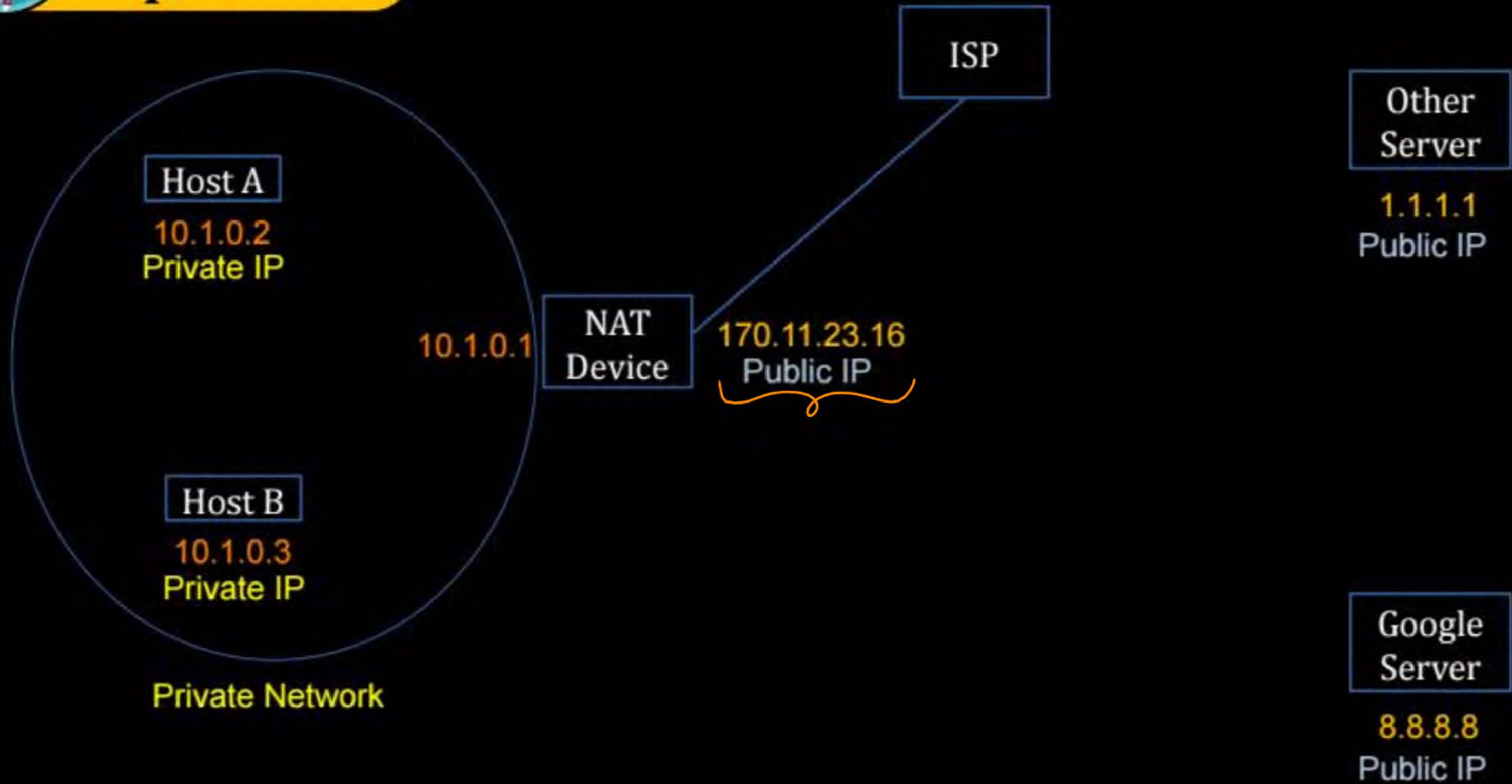
#Q. Which of the following fields (is/are) modified in the IP header of a packet going out of a network address translation (NAT) device from an internal network to an external network?

- ☒ A Header Checksum
- ☒ B Source IP
- ☐ C Destination IP
- ☐ D Total Length

Ans: A & B



Topic : NAT





Topic : NAT Table

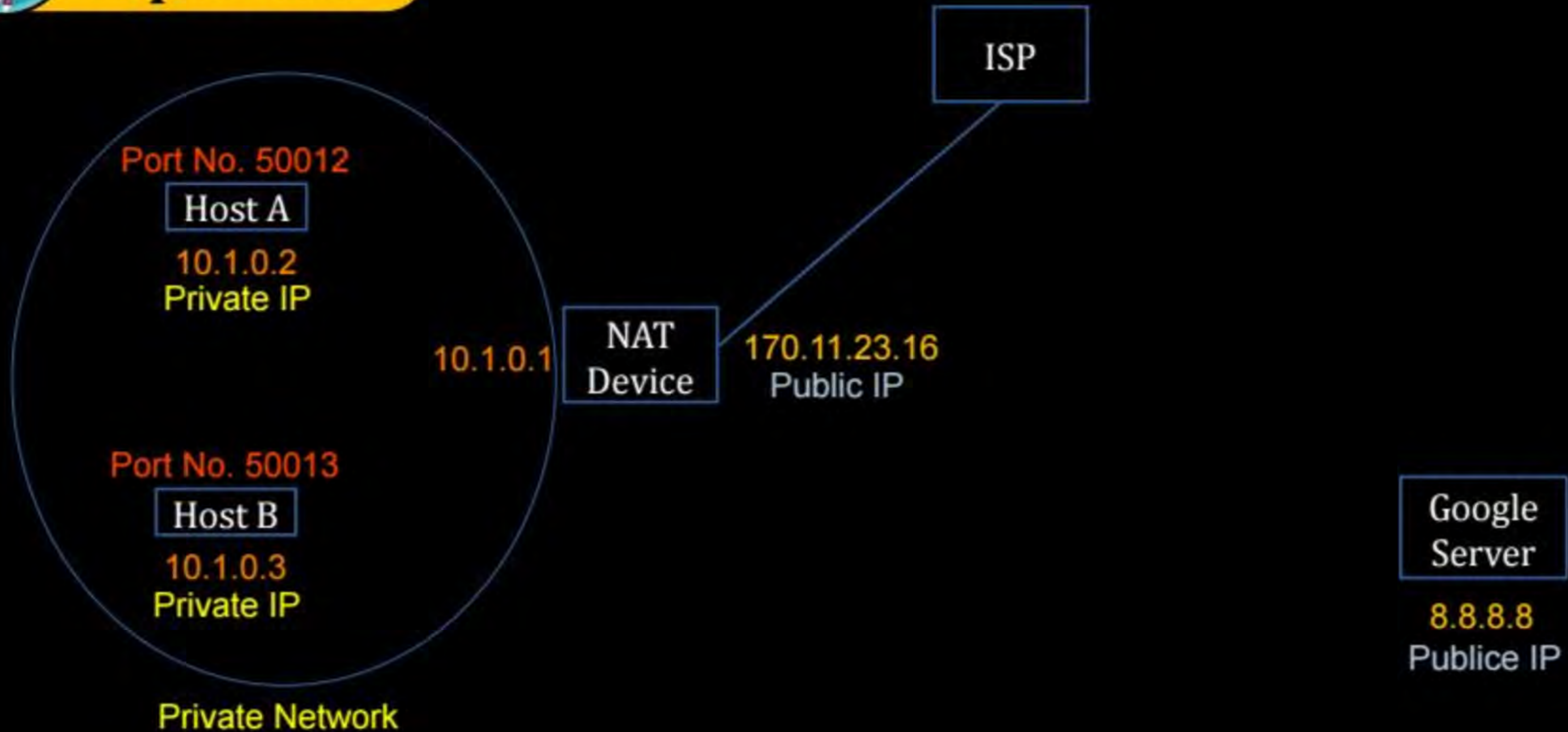


→ NAT device maintain,
NAT table for address translation of incoming datagram

Local Private IPv4 Address [Source IP Add.]	Global Public IPv4 Address [Destination IP]
10.1.0.2 [H _A]	1.1.1.1 ✓
10.1.0.3 [H _B]	8.8.8.8 ✓



Topic : NAT





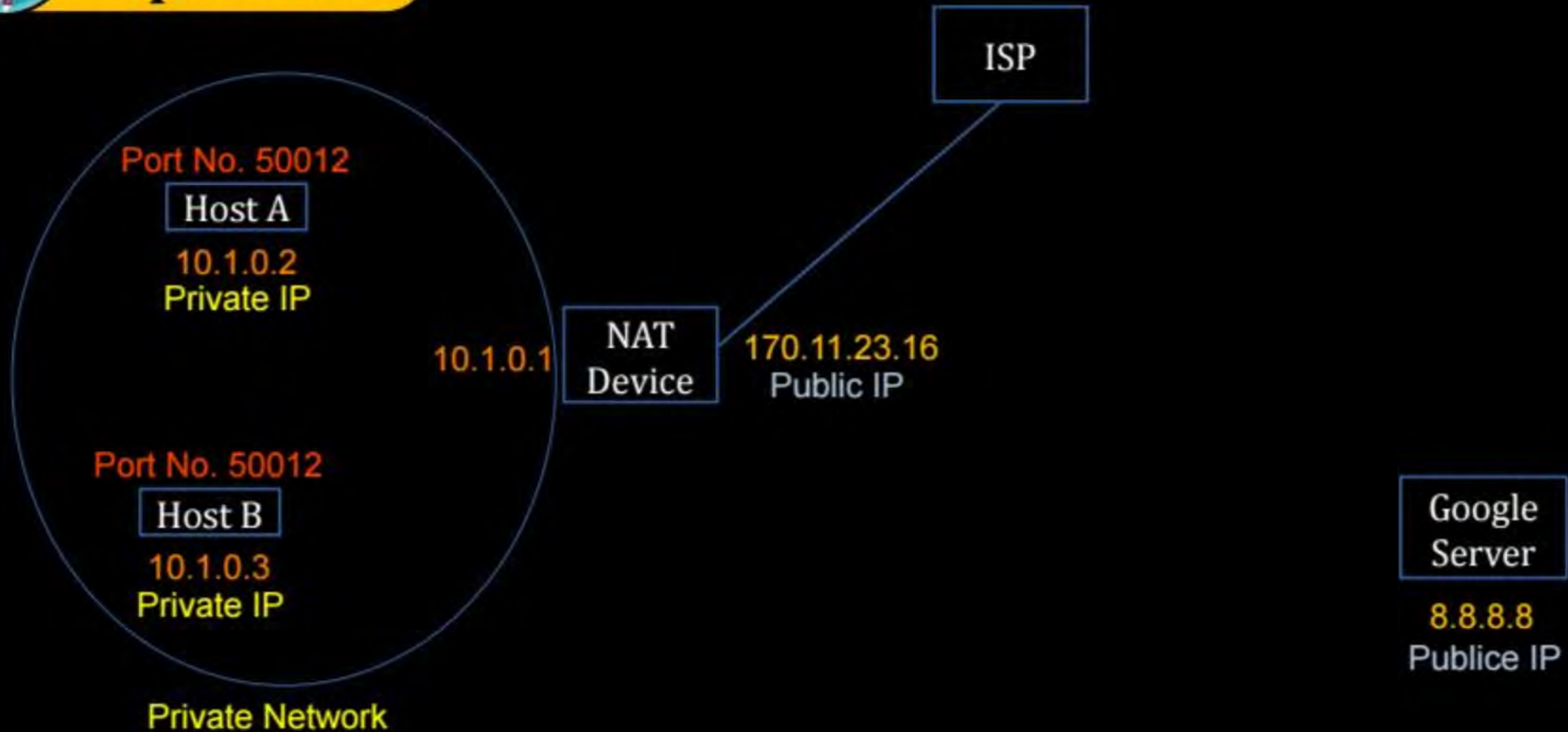
Topic : NAT Table



Local Private IPv4 Address [Source IP Add.]	Global Public IPv4 Address [Destination IP]	Source Port Number
10.1.0.2 [HA]	8.8.8.8	[50012] [HA]
10.1.0.3 [HB]	8.8.8.8	[50013] [HB]



Topic : NAT





Topic : NAT Table



Local Private IPv4 Address [Source IP Add.]	Global Public IPv4 Address [Destination IP]	Source Port Number	Modified Source Port Number
10.1.0.2 [HA]	8.8.8.8	50012	50012
10.1.0.3 [HB]	8.8.8.8	50012	[50020]

TTL

24Hr

24Hr



Topic : NAT Device

⇒ IPAT



→ As per requirement, NAT device can modify
Source Port Number field for outgoing packet
and Destination Port Number field for incoming packet

Example :-

[MCQ]



#Q. Which NAT type where single public IP address map to multiple private IP addresses using different port numbers?

- ☐ A Static NAT
- ☐ B Dynamic NAT
- ☒ C PAT $\Rightarrow$ [Port Address Translation]
- ☒ D None of the above

NAT
Overloading



Topic : NAT Device



→ NAT device update address field of every outgoing and incoming IP packet

→ Mapping internal IP addresses and ports to external IP addresses and ports



Topic : Traffic Shaping



→ Congestion control at router
[Congestion occur due to bursty traffic]

→ Two traffic shaping algorithm :-

- ✓ 1. Leaky Bucket Algorithm
2. Token Bucket Algorithm

} ⇒ Exit point
① Computer Network Layer
② Router
③ ISP Router



Topic : Leaky Bucket Algorithm



→ Convert unregulated flow into regulated flow

→ Outflow is at a constant rate (Fixed Rate)

Input Rate = Sum of all Input

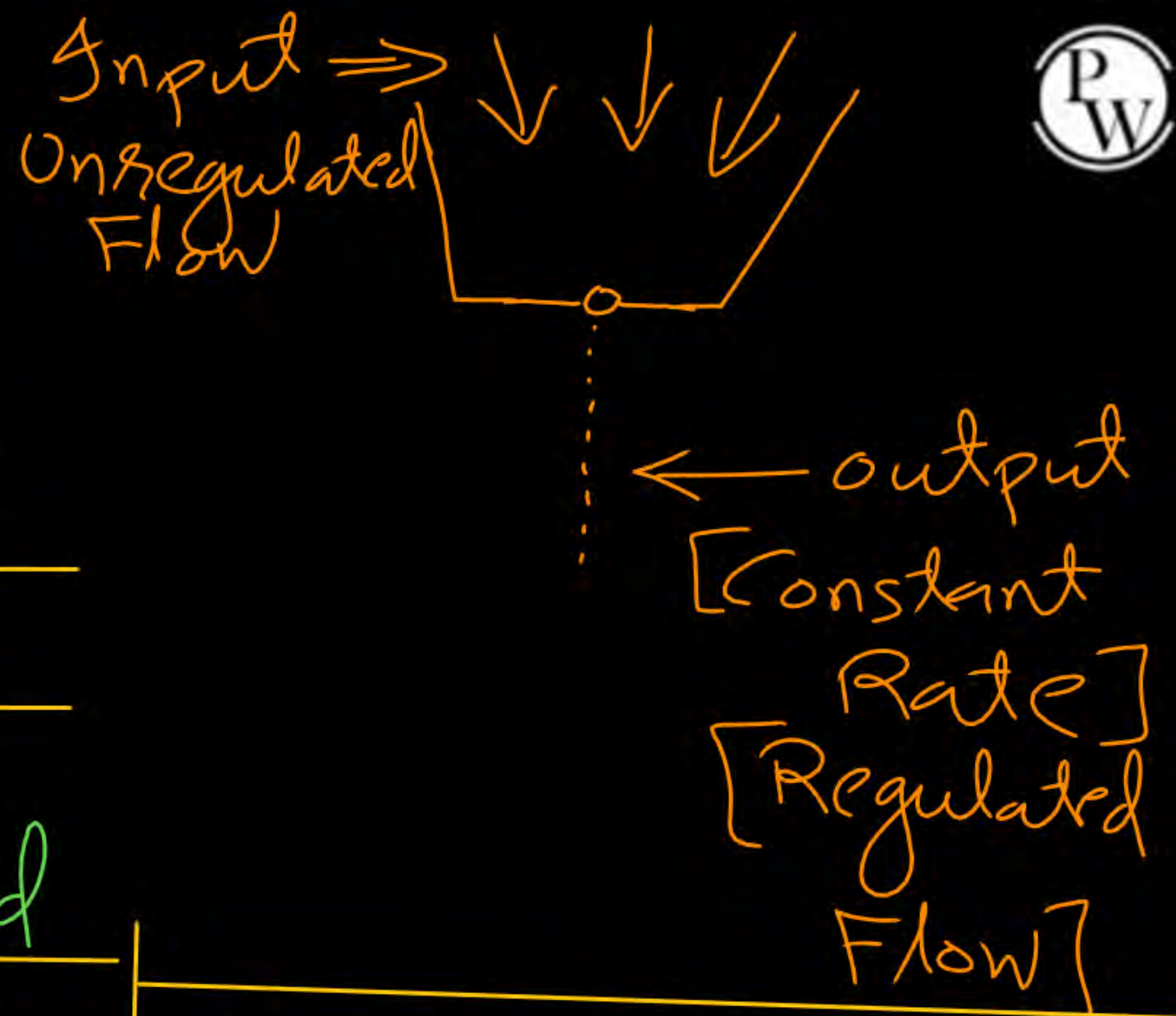
if (Input Rate $\leq$ output Rate)

→ No Bucket (Buffer) required

if (Input Rate $>$ output Rate)

→ Need of Bucket (Buffer)

* Overflow can be there.



=> In Leaky bucket burst traffic handle by bucket (buffer)

if (bucket is empty)

Output Rate = $\text{Min}[\text{Input Rate}, \text{Output Rate}]$



if (bucket is not empty and $\text{input Rate} < \text{output rate}$)
 $\Rightarrow$ Output Rate is maintain till bucket get empty.



Topic : Token Bucket Algorithm

$$\text{ISP} \rightarrow \frac{R \text{ Mbps}}{R < M}$$



→ Sometimes needed to speedup the output rate for a while

C = Capacity of token bucket (in bytes) [Max^m Limit = c byte]

R = Token arrival rate (in bytes per sec) (bits/sec) (Token generate)

M = Maximum output rate (in bytes per sec) (bits/sec)

[S = Duration for maximum output rate permissible (in sec) in case of burst traffic

$$[C + R * S] = M * S$$

$$S = \frac{C}{(M - R)}$$

→ Excess token can cause overflow

If (input Rate $< R$)

→ Excess token are stored in bucket;
→ output rate = input rate

If (input Rate = R)

→ Output rate = R (maintain)

If (input rate $> R$)

→ burst traffic

If token bucket is empty

⇒ output rate = R

If token bucket is not empty

⇒ output rate can be M for S duration



[MCQ] [GATE-CS-2008, 2-Marks]

11SC



#Q. A computer on a 10Mbps network is regulated by a token bucket. The token bucket is filled at a rate of 2Mbps. It is initially filled to capacity with 16 Megabits. What is the maximum duration for which the computer can transmit at the full 10Mbps?

- ☐ A 1.6 seconds
- ☒ B 2 seconds
- ☐ C 5 seconds
- ☐ D 8 seconds

Ans: B

Solution :-

$$\underline{C} = \underline{16 \text{ Mb (Megabits)}} \text{ (Full)}$$

$$\underline{M} = \underline{10 \text{ Mbps}}$$

$$\underline{R} = \underline{2 \text{ Mbps}}$$

$$\underline{S} = \underline{\text{Maximum duration for which}} \\ \underline{\text{the computer can transmit at the full capacity}} \text{ (max}^m \text{ output rate)}$$

$$S = \frac{C}{(M - R)} = \frac{16 \text{ Mb}}{(10 \text{ Mbps} - 2 \text{ Mbps})} = \frac{16 \text{ Mb}}{8 \text{ Mb/sec}} = 2 \text{ sec}$$

[NAT] [GATE-2016, Set-1, 2-Marks] *11SC, H.W.*



#Q. For a host machine that uses the token bucket algorithm for congestion control, the token bucket has a capacity of 1 megabyte and the maximum output rate is 20 megabytes per second. Tokens arrive at a rate to sustain output at a rate of 10 megabytes per second. The token bucket is currently full and the machine needs to send 12 megabytes of data. The minimum time required to transmit the data is _____ seconds.

[MCQ] [GATE-IT-2006, 2-Marks]

IIT-KGP, H.W.



#Q. A router has two full-duplex Ethernet interfaces each operating at 100 Mb/s. Ethernet frames are at least 84 bytes long (including the Preamble and the Inter-Packet-Gap). The maximum packet processing time (in microseconds) at the router for wirespeed forwarding to be possible is _____.

- A 0.01
- B 3.36
- C 6.72
- D 8



2 mins Summary



Topic

NAT ✓

Topic

Traffic Shaping ✓

Topic

~~DHCP~~

Transport
Layer



THANK - YOU

